Development and comparison of Imaging FlowCytobot classifiers in coastal California

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Abstract

The Imaging FlowCytobot (IFCB) is emerging as a valuable new tool in phytoplankton ecology and HAB monitoring. Initial development of classifiers is time-consuming and labor intensive, requiring trained taxonomists with knowledge of local phytoplankton. However, after development and testing of a classifier for image identification, an IFCB provides round-the-clock, real-time observation of the phytoplankton community in its deployment location. In this study, IFCBs were deployed at Pier 17 in the San Francisco Bay, and at the Santa Cruz Wharf in Monterey Bay. We investigate how the performance of the classifiers developed in these specific locations compare to one developed in the Gulf of Mexico for identifying common California HAB-forming taxa, including *Pseudo-nitzschia* spp., *Alexandrium* spp., and *Dinophysis* spp. Additionally, data collected from side-by-side IFCB deployments at the Santa Cruz Wharf were used to compare the performance of the instruments to one another. This allowed us to determine how well images from two different instruments represent the phytoplankton community at a single site, and evaluate whether a classifier is inherently tied to the instrument used to build it. Additionally, image classification methodologies were compared and here we present the results of a convolutional neural network (CNN) as an alternative to well-described supervised learning methods. This work provides valuable insight into the development of a HAB-monitoring network of IFCBs along the California coast.

Keywords: harmful algae monitoring, phytoplankton, Imaging FlowCytobot, machine learning, convolutional neural network, image classifier

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Introduction

Phytoplankton play a vital role in the marine environment, and serve as indicators of ocean health. However, many species of phytoplankton produce potent toxins that can negatively impact marine life and human health. As the phytoplankton community shifts and changes regularly and rapidly in both time and space, close monitoring of phytoplankton genera is necessary. In coastal California, harmful algal bloom-producing (HAB) genera monitored by the California Department of Public Health and/or California Harmful Algal Bloom Monitoring and Alert Program (HABMAP) are *Akashiwo, Alexandrium, Ceratium, Dinophysis, Lingulodinium, Margalefidinium, Prorocentrum*, and *Pseudo-nitzschia*.

In addition to traditional phytoplankton monitoring methods, many organizations in California are embracing a new monitoring technology, the Imaging FlowCytobot (IFCB). An IFCB is an in-situ automated submersible flow cytometer that uses a combination of flow cytometric and video technology to capture high resolution images of suspended particles. Coupled with a machine learning algorithm, an IFCB can capture and identify up to 30,000 high-resolution images of phytoplankton per hour.
There are several methods commonly used to build image recognition algorithms (classifiers), and in this study we compare a variety of classifiers developed with different methods and in different locations. These data can be used to best determine allocation of resources as IFCB use becomes more common in California.

Material and Methods

IFCB deployment
In this study, the IFCBS IFCB104 and IFCB113 were used to collect data for analysis. IFCB104 was deployed beneath the Santa Cruz Wharf in Monterey Bay, and IFCB113 was deployed at Pier 17 in San Francisco Bay. At both locations, the IFCBs were positioned on the pier above the water, with water pumped to the instrument as described in Fischer *et al*. (2020).

During May and June 2020, both instruments were temporarily deployed together at the Santa Cruz Wharf for four weeks. During this simultaneous deployment the photomultiplier tube (PMT) settings for each instrument were initially the same, which resulted in a dramatic discrepancy in total cell concentration seen in the data (Fig. 1). After adjustment of IFCB113's PMT settings, total cell concentration seen by each instrument became comparable (McLane, 2018).

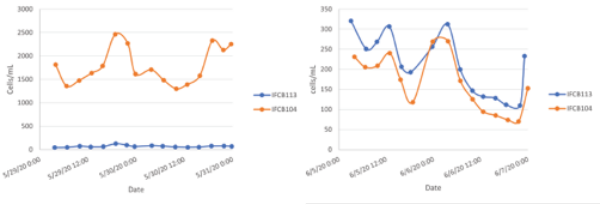


Fig. 1. Total cell counts between the instruments at the beginning of deployment with identical PMT settings (left), and after adjusting IFCB113's PMT settings to better represent the cell sizes present in Monterey Bay (right).

Random Forest classifier development
To develop the location-specific random forest classifiers, data from each location were manually classified and processed following the instructions from the IFCB

Wiki page (https://github.com/hsosik/ifcb-analysis/wiki). The Monterey Bay classifier (MBRF_24) was built with data collected from 2016-2019 and identifies 24 classes of phytoplankton (Fischer *et al*., 2020). The San Francisco Bay classifier (SFBRF_32) was built with data collected from 2018-2020, and has 32 phytoplankton classes. We also tested a classifier developed by the TOAST lab at Texas A&M (Henrichs *et al*., 2021), TOAST_RF_54, with 54 phytoplankton classes.

Convolutional Neural Network classifier development

Images from the Santa Cruz Wharf were classified using an Xception architecture depth-wise convolutional neural network (CNN) (Chollet, 2016). The model was trained using ~82,000 manually classified images from 50 distinct classes collected at the Santa Cruz Wharf. Training results indicate a validation accuracy of 0.98. Using the trained model, images from the year 2020 were classified and the highest probability class was assigned to the image. Cell concentrations were calculated as the sum of images in a given class per sample divided by the sample volume. The mean daily cell concentration for each class was subsequently calculated as the average of all daily samples.

Phytoplankton analysis with whole cell probes

Weekly water quality samples have been collected at the Santa Cruz Wharf since 2006. These data include whole cell probe (WCP) analysis for *Alexandrium catenella, Pseudo-nitzschia australis*, and *Pseudo-nitzschia multiseries* (Scholin *et al*., 1994; Miller and Scholin, 2000). These toxigenic HAB species are routinely monitored by the California Department of Public Health (https://www.cdph.ca.gov/Programs/CEH/DRSEM/Pages/EMB/Shellfish/Phytoplankton-Monitoring-Program.aspx). *Dinophysis* spp. cell concentrations were estimated using autofluorescence (Ex 470 nm, Em 515 nm) and are a potential problem in Monterey Bay (Shultz *et al*., 2009).

Results and Discussion

The initial performance of the newly developed random forest classifier SFBRF_32 is equivalent to MBRF_24, with 99.1% overall accuracy when run against the training set of images. During the side-by-side deployment at the Santa Cruz Wharf, SFBRF_32 performed similarly to MBRF_24 on data acquired from both instruments, with accuracies ranging from 94-96%. This indicates that the instrument being used for data collection is not a major factor in how a classifier performs. Additionally, it appears that SFBRF_32 and MBRF_24 yield equivalent identifications in this small subset of data.

To further explore the cross-application of classifiers, we compared the community composition of major phytoplankton groups (HABMAP HAB genera, diatoms, dinoflagellates, and other) as interpreted by the four classifiers over an 8-month time series in 2020 (Fig. 2). A paired t-test on log transformed data showed the HABMAP HAB group from SCW_CNN_50 was statistically similar to MBRF_24, while the other pairings were not. However, Pearson's r correlation revealed that the HABMAP HAB groups for SFBRF_32, SCW_CNN_50, and TOAST_RF_54 were all correlated well with the results produced by MBRF_24. These findings point to the need for regionally tuned classifiers

when trying to assess community structure and ecology, but still show some potential for non-optimized classifiers for evaluating HAB genera.

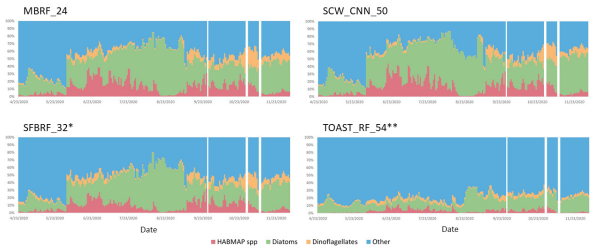


Fig. 2. Percent community composition in 2020 at the SCW based on daily averaged cell concentrations for four classifiers. Classes were grouped by CalHABMAP genera (*Akashiwo, Alexandrium, Ceratium, Dinophysis, Lingulodinium, Margalefidinium, Prorocentrum,* and *Pseudo-nitzschia*), diatoms, dinoflagellates, and other (includes unclassified).

The Pearson's r correlation confirmed MBRF_24 had strong relationships with the other classifiers for *Alexandrium, Dinophysis,* and *Pseudo-nitzschia*, but like the HABMAP HAB species group data, the classifiers were statistically different for these three genera (paired t-test, $p < 0.05$). HABs are generally defined by a cell concentration threshold and the IFCB and WCP cell concentration data were categorized as "bloom" or "not bloom" based on the threshold for each genus (Fig. 3). The bubble plots and a chi-square test confirm ($p < 0.05$) that binarized concentrations from all four classifiers are similar to the WCP data for *Dinophysis* and *Pseudo-nitzschia*. The binarized *Alexandrium* data from the classifiers are statistically different from the WCP results, but the three classifiers are statistically similar to each other. Indicating that, while not perfect or ideal, a non-tuned classifier can still provide useful information to managers about HABs.

Interestingly, the bubble plots reveal *Alexandrium* and *Dinophysis* cell concentrations were above the bloom threshold for most of 2020 at the SCW. While not surprising, as SCW was re-entering the "age of dinoflagellates" in 2018 (Fisher *et al*., 2020), it may indicate a need to re-evaluate the bloom thresholds for data collected from IFCBs.

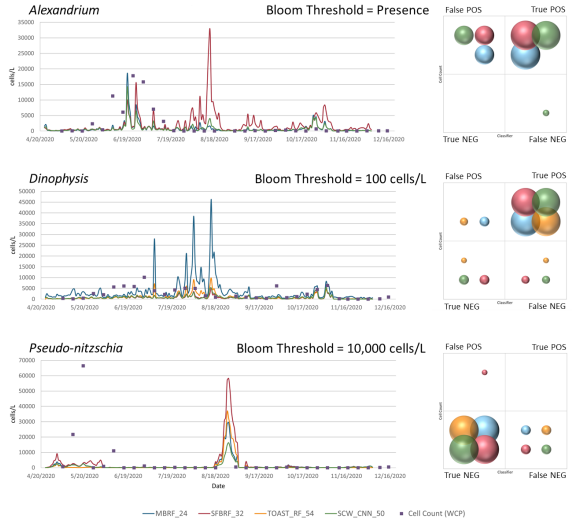


Fig. 3. Time series and quadrat plots for the three toxigenic CA HAB genera. Colors (lines and bubbles) represent the four classifiers. Cell counts are weekly using whole cell probes (WCP). Quadrat plots compare daily averaged classifier results to real-time WCP counts based on regulatory bloom thresholds for each genus. Bubble size represents the number of instances (bloom or no bloom) in each category. No bubble = no occurrence.

The *Pseudo-nitzschia* bubble plot alarmingly implies that classifiers and IFCBs are not able to identify *Pseudo-nitzschia* blooms. However, the *Pseudo-nitzschia* cell concentrations represented by the IFCB images are generally an underestimate, as each image of *Pseudo-nitzschia* is at least

two cells (end to end chains are a diagnostic characteristic). Given the wealth of data an IFCB provides, it may be worth evaluating standard bloom threshold values which were originally based on point samples.

In the next year the State of California will have 10 IFCBs deployed along its coastline. Only two of those sites have classifiers tuned to their region (Monterey Bay and San Francisco Bay) and 1 additional site has an image library large enough to build (or tune) a classifier (San Diego). The results from this study show that "tuned" IFCBs will collect equivalent data sets that can be comparable across sites. Results also indicate that a region-specific random forest classifier is as powerful as its equivalently-trained CNN counterpart. As these regional models are being applied to other locations, validation becomes an important step to ensuring any algorithm is providing accurate phytoplankton community structure data.

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References

Chollet, F. (2016). IEEE Xplore 7, 1251-1258.

Fischer, A., Hayashi, K., McGaraghan, A., Kudela, R.M. (2020). Limnol. Oceanogr. 65, 2125-2141

Henrichs, D., Angles, S., Gaonkar, C., Campbell, L. (2021). Environ. Sci. Pollut. Res. 28, 28544-28555.

McLane Research Laboratories (2018). Imaging FlowCytobot User Manual, Rev 18.F.01.

Miller, P.E. and Scholin, C.A. (2000). J. Phycol. 36: 238-250.

Scholin, C.A, Herzog, M., Sogin, M., Anderson, D.M. (1994). J. Phycol. 30, 999-1011

Shultz, D., Campbell, L., Kudela, R.M. (2019). Harmful Algae 88, 101641.

Sosik, H.M. and Olson, R.J. (2007). Limnol. Oceanog. Meth. 5, 204-216.

TOAST: https://sites.google.com/tamu.edu/phytolab/toast